

Supply-Side Teacher Forecasting Framework

Primary and Secondary Education in Tanzania

Prepared for:

Ministry of Education, Science and Technology (MOEST)

Prime Minister's Office - Regional Administration and Local Government (PMO-RALG)

A technical framework for workforce planning, budgeting, and teacher deployment.

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Tanzania has expanded access to basic and secondary education through fee-free education, curriculum reforms, and school expansion. These gains have increased **demand for teachers**, especially in areas with growing enrolment.

Persistent constraints include fiscal pressures, uneven deployment, limited training capacity in some subjects, and attrition.

Acute shortages remain visible in **STEM**, languages, special needs education, and remote localities.

Why forecasting?

A regular forecasting system enables MOEST and PMO-RALG to shift from reactive staffing decisions to **proactive workforce planning**.

Purpose

To provide a **clear, repeatable, and transparent** process for forecasting the supply of primary and secondary teachers in Tanzania.

What the framework supports

- Recruitment planning
- Budgeting and wage-bill discussions
- Teacher deployment and post allocation
- Scholarship and training decisions

What it covers

- Current stock of teachers
- New graduates and re-entrants
- Retirements, mortality, and attrition
- Disaggregation by level, subject, gender, and region

Institutional Arrangements and Governance

MOEST: policy direction, national standards, technical oversight, and use of forecasts for teacher education and sector planning.

PMO-RALG: deployment, LGA coordination, stock and attrition data, and local interpretation of forecast results.

Teacher education institutions, TCU, NACTVET: intake, completion, and subject specialization data.

Ministry of Finance, NBS, development partners: wage-bill implications, demographic projections, and technical support.

Recommended coordination mechanism

A joint **Teacher Workforce Planning Technical Working Group** should review data, validate assumptions, agree on scenarios, and sign off on forecast outputs.

Core approach

Stock-and-flow model as the main forecasting backbone

Trend-based and ratio-based projections where detailed data are incomplete

Scenario analysis to test plausible alternative futures

Structured expert judgment to refine assumptions

Why this approach?

It is **transparent**, requires relatively little data, supports **data-driven decisions**, and can be institutionalized while maintaining attention to **equity, gender, and inclusion**.

Core Stock-Flow Identity

Dynamic accounting relationship

$$\text{Stock}_{t+1} = \text{Stock}_t + \text{Inflows}_t - \text{Outflows}_t$$

Stock

Teachers in post
Actively available to
teach

Inflows

New graduates
recruited
Re-entries
Transfers in

Outflows

Retirements
Mortality
Attrition and exits

Interpretation

The model updates the teacher workforce **year by year**, making it suitable for annual planning and budgeting cycles.

Inflows and Outflows

Inflows

$$H_t = \text{Intake}_{tk} \cdot c$$

Training intake drives future graduate supply

c : completion rate

: recruitment or absorption rate

Re-entries add a smaller but relevant source of supply

Outflows

$$\text{Outflow}_t = \text{Ret}_t + \text{Mort}_t + \text{Attr}_t$$

Retirements are relatively predictable

Mortality is usually small but non-negligible

Attrition is the **most policy-sensitive** component

Scenarios and Validation

Scenario design

Baseline: continuation of current trends

Optimistic: stronger training capacity and retention

Pessimistic: fiscal tightening or higher attrition

Validation

Back-testing with historical data

Mean Absolute Error (MAE)

Root Mean Square Error (RMSE)

Mean Absolute Percentage Error (MAPE)

Planning value

Scenario analysis supports **risk-aware teacher workforce planning** rather than dependence on a single point forecast.

Implementation Roadmap

① Phase 1: Pilot and prepare

Map data sources, assess quality, integrate historical data

Pilot a basic stock-flow model for selected regions or subjects

② Phase 2: Scale up

Extend to all regions and subject groups

Align forecasting cycle with budgeting and recruitment decisions

③ Phase 3: Consolidate and enhance

Introduce more advanced statistical modules where data permit

Integrate supply and demand forecasting for fuller workforce planning

Conclusion and Policy Message

Core message

A practical supply-side forecasting framework allows Tanzania's education system to move from **reactive staffing responses** toward **strategic teacher workforce management**.

It strengthens recruitment planning, deployment, and budgeting.

It makes assumptions explicit and testable.

It improves attention to equity, gender balance, and underserved areas.

It can begin with existing tools and evolve as data quality improves.

Every learner should have access to a qualified teacher.